

Supplementary Information

An open-source, 3D-printed auger powder doser designed with generative artificial intelligence

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S1 Bill of materials

Table S1 lists the complete per-channel bill of materials for the single-channel module, plus system-shared items (one per platform regardless of channel count). Prices are single-unit USD as of 2026-Q2 and should be reconfirmed at order time; vendor links and mirrored vendor design files are maintained in `hardware/` in the repository.

Table S1: Bill of materials. Printed parts are excluded (about 350 g PLA per channel, <\$10), as is the gravimetric-feedback instrument (A&D HR-100A analytical balance, 0.1 mg readability), shared laboratory equipment that most labs already own.

Part	Qty	Price (USD)	Source
<i>Per channel</i>			
NEMA-11 bipolar stepper, 0.67 A, 10 N·cm (11HS18-0674S)	1	14	StepperOnline
Pololu Tic T500 stepper controller	1	33	Pololu #3135
JF-0530B 5 V push-pull solenoid (tap actuator)	1	7.50	Adafruit #412
DRV8871 H-bridge breakout (solenoid drive)	1	7.50	Adafruit #3190
ERM coin vibration motor, 10 mm	1	1.95	Adafruit #1201
DRV2605L haptic driver breakout (I ² C)	1	7.95	Adafruit #2305
MG996R-class metal-gear servo (tilt)	1	12	generic
6805ZZ bearing (auger support)	1	3	generic
5 mm shaft coupler, fasteners (M3/M5), bulk capacitors	—	10	generic
<i>System-shared (one per platform)</i>			
Raspberry Pi Pico W	1	6	Raspberry Pi
MAX3232 RS-232 transceiver breakout (balance interface)	1	3	generic
Mean Well GST60A12-P1J 12 V/5 A supply + cord + pigtail	1	26	Digi-Key
Pololu D24V22F5 5 V buck regulator	1	19	Pololu #2858
Pololu 33 V shunt regulator (back-EMF clamp)	1	15	Pololu #3776
Breadboard/protoboard, headers, wiring	—	10	generic
Total, single-channel platform		≈175	

S2 Construction guide

A step-by-step illustrated build guide is maintained in the repository (<https://github.com/vertical-cloud-lab/powder-doser>) alongside the parametric CAD sources and STLs for every part. In outline:

1. **Print** the auger, gear pair, bracket, tap collar (two clamp halves), mounting plate, hinged baseplate, and servo sector in PLA (0.2 mm layers, 4 perimeters, 25% infill, no supports; print the auger vertically, helix up).

2. **Auger drive:** press the 6805ZZ bearing into the bracket, seat the auger, mesh the printed gear pair, and bolt the NEMA-11 stepper to the front face of the mounting plate.
3. **Tap collar:** clamp the two collar halves around the auger tube; mount the solenoid to the collar flange (2×M3) so the plunger strikes the anvil pad, and adhere the ERM motor to the opposite pad.
4. **Tilt stage:** join the mounting plate to the baseplate through the side hinges (printed pins), engage the servo sector, and confirm the exit nozzle lies on the hinge axis.
5. **Electronics:** wire the Pico W, Tic T500 (UART), DRV8871, DRV2605L (I²C), MAX3232 (RS-232 to the HR-100A balance), and servo per the KiCad schematic in `hardware/test-module/`; flash the MicroPython firmware.
6. **Checkout:** run the firmware self-test (rotate, tap, vibrate, tilt, tare/read), then a per-powder calibration sweep.

S3 Exit-nozzle variants

Four printed exit-nozzle geometries were produced for dispense testing (Fig. S1); the dispensing campaign will report flow-initiation and dribble behaviour per nozzle and per powder.

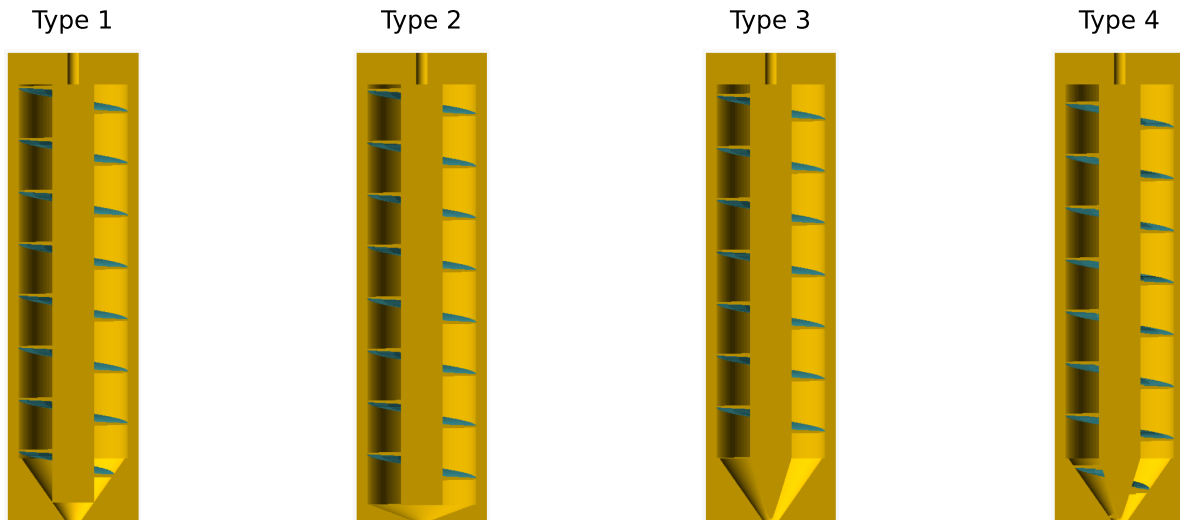


Figure S1: Cross-sections of the four printed exit-nozzle variants (Types 1–4) of the test auger, differing in exit constriction and chamfer geometry.

S4 AI usage log

In line with Digital Discovery’s guidance on LLM use, all AI interactions are preserved verbatim and publicly:

- **CAD/code generation:** every agent session (GitHub Copilot coding agent; Claude-family models, April–June 2026) is recorded as a pull request containing the prompt (issue text and review comments), the generated CAD code, rendered views, and the human review thread.
- **Design reviews and literature queries:** Edison Scientific query scripts, task identifiers, raw answers, and extracted references are archived under `paper/background/` on the corresponding branches; the consolidated, validated bibliography with per-entry verification status is at `paper/references_validation.md`.
- **Text-to-CAD evaluations:** CADSmith run artifacts and the Zoo Design Studio (Zookeeper agent) session transcripts and best-practice notes are preserved on their respective branches.
- **Qualitative observations:** the running team journal of AI behaviour (good and bad) is public in the repository’s discussion #39.